CPSC 581: Project 1

Team Members	
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Write Up:

For our design process, because of the limitations of only having two group members, we had the freedom to create a more linear design that was more directly interactive between the two of us. We combined what the TA showed us as an example with the ideas presented. The ideas focused on a central figure with buttons that interacted with it to react in ways the designer intended to represent themselves. This sparked our idea of a central figure and two buttons that interact in similar ways. As a group, we discussed what we were each passionate about to generate ideas that represented our interests and helped us find mutual connections. The first round of discussions brought up sports as a possible topic. Since we both had a connection to sports, we decided it would be a good starting point. We then narrowed our choice to soccer because it was a good starting point for exploring different ways we could interact with a soccer ball. We considered ways to interact with a soccer ball, including a goalie system where one button would kick the ball, and another would block it in a video-game sense, which also connected the members (Figure 1). However, this felt more like developing a real video game than showing the connections between the members.

After discussing the possibilities of two buttons in a “game like” sense, we considered having one button to kick the ball and another to fix it. This shows the connection between both members, highlighting their differing interests and how they interact with each other. Later, we refined the idea into a more niche interaction: one button kicks the ball toward a wall, and the other button fixes the wall and the building's structure in a video game-like scenario. This was the final design we settled on to show a more DIY-style interaction that one member leaned toward (Figure 2). This version at the time had a focus on the repair side of DIY activities, where we would repair the ball as it wore down by pressing the “kick” button, and the “repair” button would fix the ball. The final application we settled on revolved around kicking the ball towards a wall, slowly breaking its structure. Once the ball depleted the wall’s health, the structure would collapse, prompting the user to fix the building with DIY skills, or the user could click the button to repair the wall before it broke (Figure 3).

Regarding AI Citation, we used the Cursor IDE integrated with the Grok 4.6 Medium model framework. Generative AI was leveraged to scaffold our structural HTML file framework, apply mobile-first styling via Tailwind CSS utility classes, and compute CSS transition positioning logic for cross-device web responsiveness. The primary prompt provided was the following: "Help with the creation of the framework regarding a highly interactive, responsive single-page web application titled 'A Family of Buttons'. The app features exactly two primary interactive buttons representing real-life hobbies: Soccer ('Kick Ball') and DIY Projects ('Repair Wall'). The layout must feel entirely natural on iPhone, iPad, laptops, and wide monitors using Tailwind breakpoints and comfortable mobile touch targets." While the

AI helped with raw code markup, the underlying conceptual metaphor of damage/impact vs restoration, design asset parameters, and responsive touch-target criteria were engineered, verified, and directed entirely by us.

Figure 1. First sketches for button interaction.

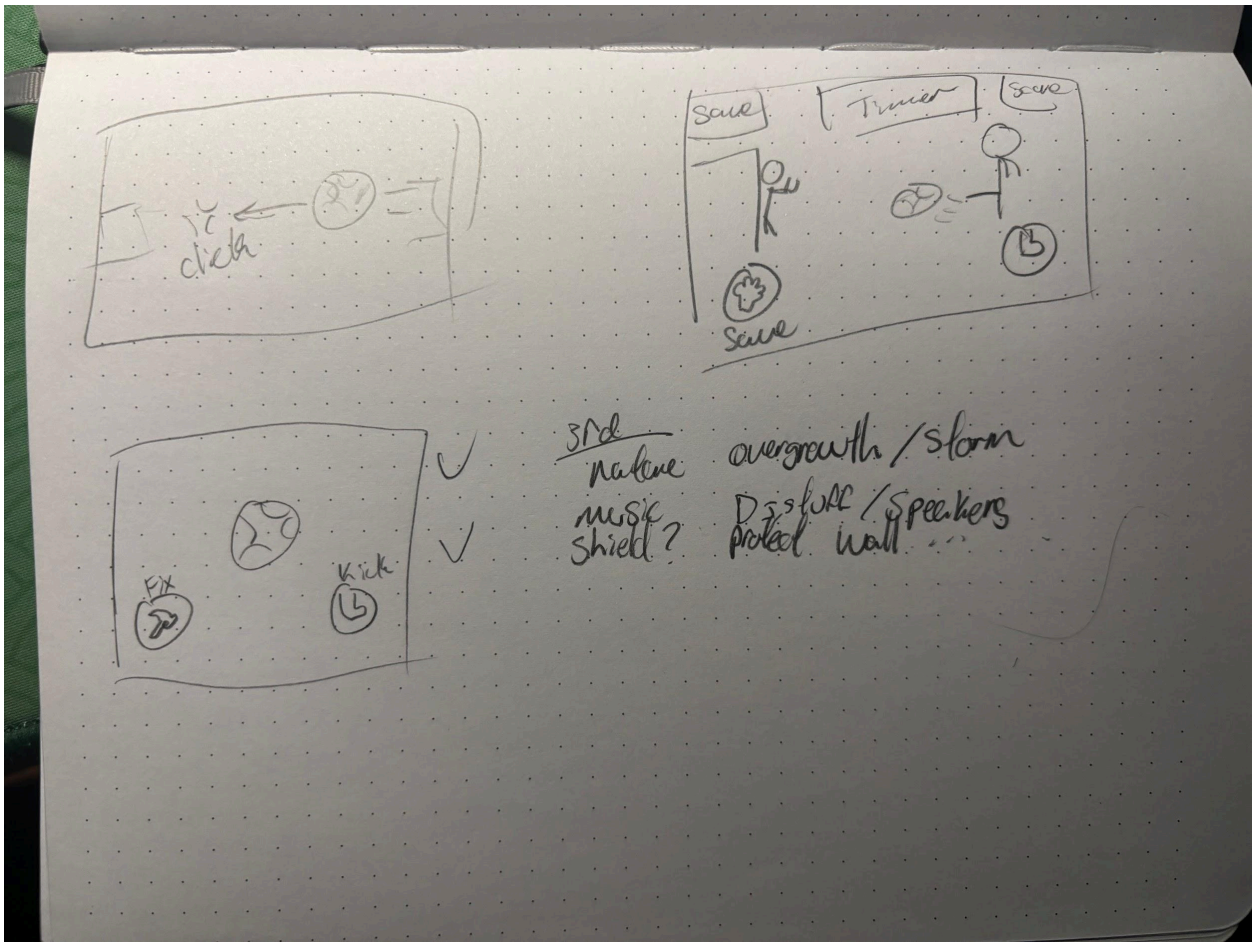


Figure 2: Revision into idea of building and breaking something

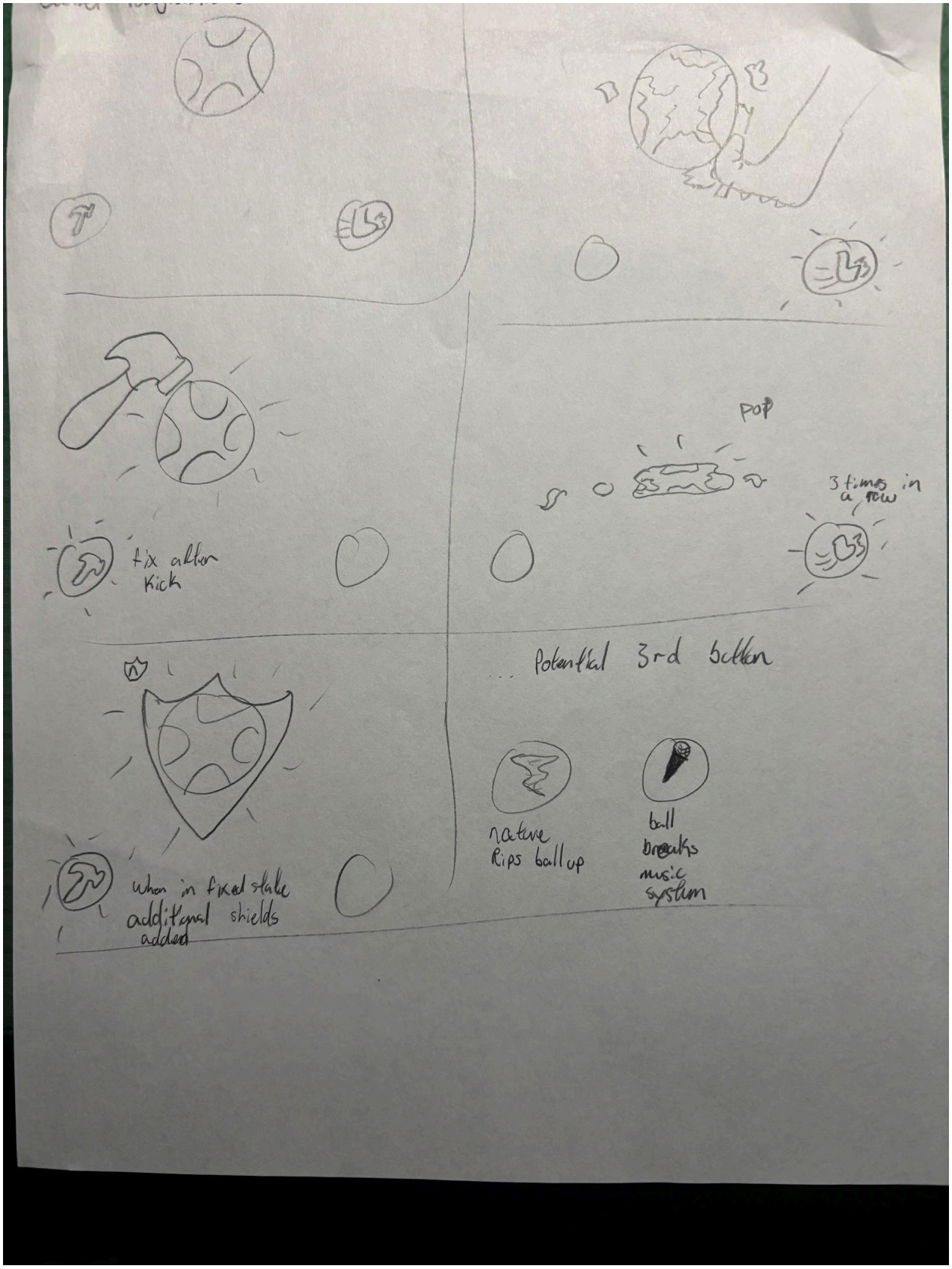


Figure 3: Final Sketches.

